

Customer Churn Analysis - Music Streaming Subscription Platform

SQL + Data Visualization Case Study

Executive Summary:

This project looks at customer churn behavior in a subscription based music streaming platform. Using SQL and Google Sheets, I examined user engagement and subscription types. I also looked at behavioral patterns to identify key drivers of churn and potential high risk customers.

The analysis reveals that churn is strongly influenced by user engagement levels and subscription tier, with free users showing higher churn rates compared to paid users.

Problem Statement:

The platform is experiencing a high customer churn rate, reducing long term user retention and revenue potential. The goal of this analysis is to identify:

- Overall churn rate
- High-risk customer segments
- Behavioral patterns associated with churn
- Actionable strategies to improve retention

Dataset:

This project uses a publicly available synthetic dataset from Kaggle:

Streaming Subscription Churn Dataset

<https://www.kaggle.com/datasets/raghunandan9605/streaming-subscription-churn-dataset>

The dataset simulates user behavior on a music streaming platform, including subscription type, weekly usage, session behavior, and churn status.

The music streaming platform is comparable to Spotify, Apple Music etc.

Tools Used:

- SQL (data analysis & aggregation)
- Google Sheets (dashboard creation)

Methodology and Approach:

1. Data Exploration

- Reviewed key variables such as user engagement (weekly hours, session length), subscription type, and churn status.

2. SQL Analysis

- Used SQL to calculate churn rates, compare subscription types, and summarize behavioral differences between churned and retained users.

3. Behavioral Segmentation

- Compared churned vs non churned users based on engagement metrics like weekly usage, skip rate, and playlist activity.

4. High risk identification

- Flagged users with low engagement and high skip rates as high risk churn candidates.

5. Visualization

- Summarized insights using Google Sheets dashboards to highlight churn patterns and user behavior trends.

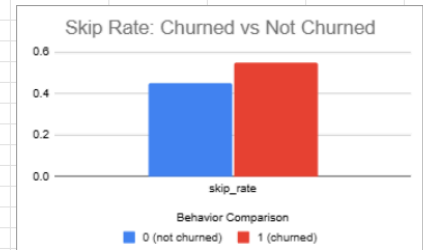
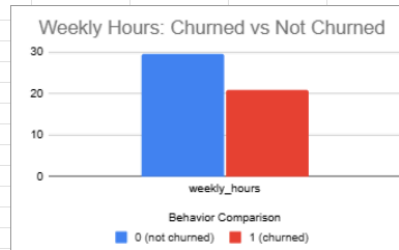
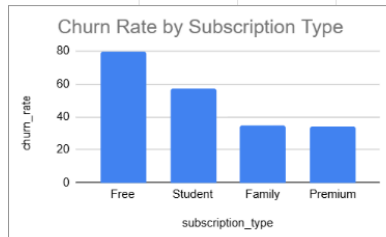
Key Insights:

- The churn rate is 51.34%, indicating that more than half of users leave the platform.
- Free users: 79.39% churn
- Premium users: 33.87% churn
- Subscription type is the strongest predictor of churn.
- Churned users have lower weekly listening hours (20.82 vs 29.52)
- Churned users have a little higher skip rates (0.55 vs 0.45)
- Playlist creation does not significantly impact churn behavior.

Visualization:

Customer Churn Analysis - Music Streaming Platform: Dashboard

Overall Churn Rate **51.34%**



8062
High-risk users

Recommendations:

- Improve early user engagement through onboarding and personalization
- Enhance recommendation quality to reduce skip rates
- Target high risk users (8,062 segment) with retention campaigns

Conclusion:

Churn is mostly driven by user engagement rather than pricing alone. The data shows that increasing weekly usage and improving content relevance are the most effective strategies to reduce churn and improve retention.